

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Regular / Supplementary Examinations Jan / Feb-2025

Industrial Engineering and Management**WORK SYSTEMS DESIGN**

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

M BT CO

1	1.1	Illustrate the Pyramidal Structure of the work.	02	2	1
	1.2	Distinguish between internal element and external element in cycle time analysis.	02	4	1
	1.3	Enumerate the categories of worker machine systems based on the levels of operator attention required.	02	1	2
	1.4	Differentiate between snap back and cumulative timing methods.	02	4	2
	1.5	What is lean production?	02	2	3
	1.6	Why is an allowance added to the normal time to complete the standard time?	02	2	2
	1.7	Identify the broad categories of factors causing fatigue in workers.	02	2	4
	1.8	List any four techniques of Work Measurement.	02	1	2
	1.9	In the context of lean manufacturing what is production leveling?	02	2	3
	1.10	In lean manufacturing, _____ refers to waste or non-value-adding activities, while _____ refers to overburdening people or equipment.	02	2	3

PART-B

2	a	Distinguish between Production and Productivity. Also justify the rationale for considering labor productivity as a measure of organizational efficiency and effectiveness.	08	4	1
	b	The normal time to perform a repetitive manual task is 4.25 minutes. In addition, an irregular work element whose normal time is 1.75 minutes must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance is 16%. Determine the standard time per piece. How many work units are produced in an 8-hour shift at standard performance? Determine the anticipated amount of time worked and the amount of time lost per 8-hour shift that corresponds to the PFD allowance factor of 16%.			
3	a	Review the applicability of three categories of Charting / diagrammatic techniques used by industrial engineers for collecting, displaying and analyzing data on a given work system.	08	2	1
	b	Provide an overview on any four techniques of Methods Engineering.	08	2	1

OR

4	a	Enumerate and provide a brief note on the principles of motion economy.	09	2	1
	b	Review the phases involved in the systematic approach that can be used to accomplish a methods engineering study.	07	2	1

5	a	What is a Work Element? Mention at least six guidelines for defining work elements in direct time study. Supplement with appropriate explanation and examples.	08	3	2
	b	The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element <i>c</i> is a machine controlled and the time is constant. Elements <i>a, b, d, e</i> and <i>f</i> are operator controlled and external to machine cycle and they are performance rated at 90%. If the machine allowance is 12% and worker <i>PFD</i> allowance is 8%, determine the normal time and standard time for the work cycle.			

Element	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
Observed Time(min)	0.12	0.28	0.78	1.04	1.45	1.80

OR

6	a	What is meant by <i>PMTS</i> ? Distinguish between First level <i>PMTS</i> and Higher level <i>PMTS</i> on the basis of at least five characteristics.	08	2	2
	b	Analyse the causes for variability in the observed work element times that occur from one work cycle to another work cycle.	08	4	2

7	a	Enumerate the seven wastes as advocated by Taiichi Ohno in Lean Production philosophy. Explain any four in detail.	10	3	3
	b	Distinguish between Push System of production control and Pull System of production control. Support your answer with an appropriate illustration.	06	4	3

OR

8	a	In Toyota Production System a standardized work procedure for a given task has three components. What are they? Explain any two in detail.	08	2	3
	b	Explain the role of Total Productive Maintenance (<i>TPM</i>) in automation with examples.	08	2	3

9	a	Employees in an office environment report eye strain, fatigue, and discomfort. Analyse the physical work environment and recommend improvements.	08	3	4
	b	What is ergonomics and explain its purpose in production systems. Discuss how ergonomics and human factors are connected to engineering.	08	2	4

OR

10	a	Highlight on the concept of anthropometry and its importance in designing for the human body, with examples.	08	2	4
	b	Describe how mechanics can be applied to the human body to enhance safety and efficiency in the workplace.	08	2	4

**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	21 st October 2024	Maximum Marks	10 + 50
Course Code	IM233AI	Duration	120 Mins
Sem	III Semester	CIE – I	
WORK SYSTEMS DESIGN			

Note:

1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is meant by the term Work?	02	2	1
2	Illustrate the Pyramidal Structure of the task.	02	3	1
3	Distinguish between methods design and methods engineering	02	2	2
4	Mention the elements of Standard Time.	02	2	1
5	What are the two common measures used in industry to assess a worker's productivity?	02	1	1

PART – B

Sl. No.	Questions	M	BT	CO
1.	Distinguish between Production and Productivity. Explain with suitable illustrations on the ways and means to enhance productivity in organizations.	10	3	CO1
2.	A work group of 10 workers in a certain month produced 1000 units of output working 8 hr/ day for 22 days in the month. What productivity measures could be used for this situation? Suppose that in the next month, the same work group produced 1200 units but there were only 20 work days in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	10	3	CO1
3.	The normal time to perform a repetitive manual task is 3.25 minutes. In addition, an irregular work element whose normal time is 1.5 minutes must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance is 12%. Determine the standard time per piece. How many work units are produced in an 8 hour shift at standard performance? Determine the anticipated amount of time worked and the amount of time lost per 8 hour shift that corresponds to the PFD allowance factor of 14%.	10	3	CO3
4.	Enumerate the steps involved in conducting a Method Study by an Industrial Engineer in organizations.	10	2	CO2
5 a.	Classify the worker –machine systems according to number of machines and workers.	05	1	CO1
b.	What does PFD stand for? Bring out the purpose of PFD allowance in determining the standard time for a task.	05	3	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	08	02	--	--	--	02	06	02	--	--	--
		Test	30	10	10	--	--	05	10	35	--	--	--



Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	2 nd December 2024	Maximum Marks	10 + 50
Course Code	IM233AI	Duration	120 Min
Sem	III Semester	CIE -II	
WORK SYSTEMS DESIGN			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO										
Part - A														
1	List any four techniques of Work Measurement.													
2	What is meant by the term, "Muda" in the context of Lean Manufacturing?	02	1	1										
3	Distinguish between work element and basic motion element.	02	2	2										
4	What is the expanded form of MOST?	02	2	1										
5	Mention the advocate of Toyota Production System.	02	1	3										
Part - B														
1	Illustrate and explain the pre requisites for determining the valid time standards.	02	1	2										
2	A time study analyst observed a repetitive work cycle performed by a worker and then studied the task using a predetermined motion time system. During the observation, the analyst made a record of the work elements. He also made a note of worker's performance and rated it as 85%. Back in the office, the analyst listed the basic motion elements for the task and retrieved from his computer the corresponding normal time values. The sum of normal times for the basic motion elements was 1.38 min. The company uses a PFD allowance factor of 15%. Determine the standard time for the task.	10	3	3										
3	The continuous timing method is used to direct time study a manual task cycle consisting of four elements: a, b, c and d. Two parts are produced each cycle. Element d is irregular element performed once in every six cycles. All elements were performance rated at 90%. The PFD allowance is 11%. Determine the normalized time for the cycle, standard time per part, the workers efficiency if the worker completes 844 parts in an 8-hour shift during which she works 7 hours and 10 minutes.	10	4	3										
<table border="1"> <thead> <tr> <th>Element</th><th>a</th><th>b</th><th>c</th><th>d</th></tr> </thead> <tbody> <tr> <td>Observed time (min)</td><td>0.35</td><td>0.60</td><td>0.86</td><td>1.46</td></tr> </tbody> </table>					Element	a	b	c	d	Observed time (min)	0.35	0.60	0.86	1.46
Element	a	b	c	d										
Observed time (min)	0.35	0.60	0.86	1.46										
4	Enumerate different categories of Allowances considered in determining the time standards. Present your understanding by providing relevant contexts in which the different categories are applied by organizations in determining standard time.	10	4	1										
5	Sketch the structure of Lean Production System. Review the key elements of the structure.	10	3	2										

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	06	02	02	10	06	04	--	--	--	--
	Test		--	--	30	20	10	20	20	--	--	--

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date		Maximum Marks	50
Course Code	IM233AI	Duration	90 Min
Sem	III Semester	Improvement Test	
WORK SYSTEMS DESIGN			

Note:

1. Answer all the Questions.

PART - A

Sl. No.	Questions	M	BT	CO
1	Sketch a typical physical work system.			
2	Distinguish between production and productivity.	02	2	CO1
3	Illustrate the pyramidal structure of a task.	02	2	CO1
4	What is meant by the term Poka-Yoke?	02	2	CO1
5	List the five steps in 5S system.	02	2	CO2
		02	1	CO2

PART - B

Sl. No.	Questions	M	BT	CO
1	A work group of 20 workers in a certain month produced 8600 units of output working 8 hr/day for 21 days. (i) What is the labor productivity ratio for this month? (ii) In the next month, the same work group produced 8000 units but there were 22 workdays in the month and the size of the work group was reduced to 14 workers. What is the labor productivity ratio for this second month? (iii) What is the productivity index using the first month as a base?	10	4	CO1
2	The standard time is to be established for a manual work cycle by direct time study. The observed time for the cycle averaged 4.80 minutes. The worker's performance was rated at 90% on all cycles observed. After eight cycles, the worker must exchange parts containers which took 1.60 minutes rated at 120%. The PFD allowance for this class of work is 15%. Determine the normal time, the standard time for the cycle and the worker's efficiency if the worker produces 123 work units during an 8 hour shift.	10	4	CO1
3	In Toyota Production System a standardized work procedure for a given task has three components. What are they? Explain any two in detail.	10	2	CO2
4	Sketch the structure of Lean Production System. Explain the features of different elements of the structure.	10	3	CO2
5	Distinguish between Push System of production control and Pull System of production control. Support your answer with an appropriate illustration.	10	3	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	06	04	--	--	02	08	--	--	--	--
	Test	Max Marks	20	30	--	--	--	10	20	20	--	--
